(D.) REDUCTION OF HORIZONTAL DIRECTIONS TO SEA LEVEL.

The resulting directions at a station, as given in the abstracts, still need a small correction to reduce them to what they would have been had the object observed upon been at the sea level. The altitude of the observing station and the distance between them does not enter into the case; the reduction is due to the circumstance that, in general, the verticals at the two stations are not in the same vertical plane. The correction * is given by $\frac{e^2}{2} \cdot \frac{h}{\rho} \sin 2\alpha \cdot \cos^2\phi$, where $e^2 = \frac{a^2 - b^2}{a^2}$ and h = altitude of the station observed upon. ρ = radius of curvature in the plane normal to the meridian, α = azimuth of the line (counted from south around by west) and ϕ = latitude of place. With $\log e^2 = \bar{7}\cdot 8305$ and $\log \rho = 6\cdot 8054$ for $\phi = 39°$ and Clarke's spheroid (of 1866), and dividing the expression by $\sin 1''$, we get for the correction in seconds and the height in metres

$$0''\cdot 000\ 066 \sin 2\alpha . h$$

This correction has been applied systematically to all measured directions of the base nets and intervening triangulation from the Salina base to the Pacific coast, but no application was made to the triangulation east of Salina base on account of the lower altitudes and consequent smallness of the correction in this part of the arc. In comparison with the magnitude of the average triangle closing error, the effect of omitting this correction, except for the higher altitudes, seems justified. About the Salina base stations the average reduction of a sight to the sea level is but $0''\cdot 02$.

The probable error of a single observation of a direction (e) is given under the list of directions at each station as a convenient index of the accuracy of the observations. When the parenthesis (*D.* and *R.*) is used, the observations were made with a direction instrument. A single observation of a direction comprises two pointings upon the signal, one with telescope direct and one with telescope reversed, and two readings (forward and backward) of each microscope, of which there are usually three, for each pointing. e is computed by formula (8) shown on page 40.

When the parenthesis (6 *D.* and 6 *R.*) is used, the observations were made with a repeating instrument and a single observation *of an angle* comprises 12 pointings upon each of two signals, 6 with telescope direct and 6 reversed, and 3 readings of the horizontal circle, at the beginning and end of the direct measure and again at the end of the reversed measure. The quantity given is the probable error of a single observation of a *direction* (not angle) and is $\frac{1}{\sqrt{2}}$ times the probable error of a single observation of an angle.

It was also computed by the formula (8) shown on page 40.

The parenthesis (3 *D.* and 3 *R.*) has a meaning analogous to (6 *D.* and 6 *R.*).

(E.) ADJUSTMENT OF BASE NETS OR OTHER TRIANGULATIONS.

The method is the same as that usually employed to satisfy the geometrical conditions of a triangulation by application of the method of least squares. For the sake of convenience the leading formulæ referring to condition observations, together

* "Geodesy," by Col. A. R. Clarke, Oxford, 1880, p. 113.

To find the linear change of length of a given arc of parallel when moved a number of seconds to the north or south of its position on the spheroid, we have: Length in metres of 1° in latitude φ—

$$P^{\circ} = 111\ 415\cdot12 \cos\varphi - 94\cdot54 \cos 3\varphi + 0\cdot12 \cos 5\varphi - \ldots\ldots$$

$$\text{and } \frac{dP}{d\varphi} = -111\ 415\cdot12 \sin\varphi + 283\cdot62 \sin 3\varphi - 0\cdot60 \sin 5\varphi$$

Hence for $\varphi = 39°$ and $d\varphi = 0''\cdot5$ and length of arc $48\frac{3}{4}°$, the change is 8·25 metres, which in comparison with the probable error in length of the geodetic connection is a small quantity.*

For the correction to the preliminary longitude we have the following data:

	Using all tabular values (A–G).	After rejection of the nine largest values.
	"	"
(a) Indiscriminate mean	$+\frac{43\cdot39}{37} = +1\cdot17$	$+\frac{7\cdot04}{28} = +0\cdot25$
(b) Mean after forming groups †	$+\frac{31\cdot80}{25} = +1\cdot27$	$+\frac{7\cdot01}{19} = +0\cdot37$

Adopting the last value, we have $\lambda_o = \lambda + 0''\cdot37 = 99°\ 16'\ 16''\cdot73$ for the final geodetic longitude of Hays station. The uncertainty of this value may be estimated as less than 1".

Respecting any change in azimuth, the values of (A–G) at the eastern and near the western parts of the arc (first and last nine stations) appear fairly well balanced; while the stations of the western or mountainous section exhibited a predominating negative sign. In order to remove this feature, the geodetic azimuths would need a diminution, which, however, is opposed to the apparent demand for the eastern part of the arc. A small change of azimuth has but a small effect upon the latitudes and hardly any upon the longitudes. Upon the whole it has been concluded to make no change in the azimuth; hence $\alpha_o = \alpha = 359°\ 44'\ 19''\cdot00$ for the line Hays to LaCrosse.

With the standard values of φ_o, λ_o, and α_o for Hays station the geodetic latitudes and longitudes of all the stations of the arc were recomputed and the definitive results of comparison of the astronomic and geodetic determinations are tabulated below.

In order to render this comparison more complete, there are also given the positions and resulting values of (A–G) when the Besselian spheroid is substituted for that of Clarke. It must be noted, however, that only a close approximation of these values could here be given, since, in strictness, the subject would demand a readjustment of the entire triangulation with the introduction of the spherical excess appertaining to the Besselian spheroid. The difference in the excess ‡ is small, even for the largest triangle "Tushar, Wheeler, Nebo," for which $\epsilon = 73''\cdot758\ 4$. The difference in ϵ is but $0''\cdot017\ 1$, or $\frac{1}{4}\frac{1}{3}\frac{1}{1}\frac{1}{4}$ of itself, and for the greater part of the arc the hundredths of a second for any angle would not be modified by the change of spheroids. There is nevertheless a small accumulated effect in the positions which may tend to introduce a twist, yet this is fully

* For Bessel's spheroid we have $P^{\circ} = 111\ 399^{m}\cdot675 \cos\varphi - 93\cdot212 \cos 3\varphi + 0\cdot116 \cos 5\varphi - \ldots\ldots$

† Groups: (3, 4, 5); (6, 7); (20, 21); (25, 26, 27, 28, 29); (30, 31, 32), and (33, 34, 35).

‡ The effect on the spherical excess of a triangle by a change of dimensions in the spheroid is given by the expression $\frac{d\epsilon}{\epsilon} = -2\frac{da}{a} + 2\cos 2\varphi\, df$; where a = equatorial radius, φ the latitude, and $2f = e^2 = \frac{a^2-b^2}{a^2}$, as given in Part I, p. 52.